

R Code

```
1  # Overfitting a data
2  # Load the built-in airquality dataset
3  data(airquality)
4
5  # Remove rows with missing values
6  air <- na.omit(airquality)
7
8  # Set seed so the same train/test split is produced
9  set.seed(123)
10
11 # Split data into training (70%) and test (30%) sets
12 train_id <- sample(1:nrow(air), size = 0.7 * nrow(air))
13
14 train <- air[train_id, ]
15 test <- air[-train_id, ]
16
17 # Create a data frame to store errors
18 results <- data.frame(
19   degree = 1:10,
20   training_error = NA,
21   test_error = NA
22 )
23
24 # Fit polynomial models from degree 1 to 10
25 for (d in 1:10) {
26
27   model <- lm(
28     Ozone ~ poly(Temp, degree = d, raw = TRUE),
29     data = train
30   )
31
32   # Predict on training data
33   train_pred <- predict(model, newdata = train)
34
35   # Predict on test data
36   test_pred <- predict(model, newdata = test)
37
38   # Calculate Mean Squared Error
39   results$training_error[d] <- mean(
40     (train$Ozone - train_pred)^2
41   )
42
43   results$test_error[d] <- mean(
44     (test$Ozone - test_pred)^2
45   )
46 }
47
48 # View the results
49 results
50
```

```
51 # CREATE THE CHART
52 library(ggplot2)
53
54 ggplot(results, aes(x = degree)) +
55   geom_line(aes(y = training_error, color = "Training error"), linewidth = 1) +
56   geom_point(aes(y = training_error, color = "Training error")) +
57   geom_line(aes(y = test_error, color = "Test error"), linewidth = 1) +
58   geom_point(aes(y = test_error, color = "Test error")) +
59   labs(
60     title = "Training vs. Test Error",
61     x = "Polynomial Degree",
62     y = "Mean Squared Error",
63     color = "Error"
64   ) +
65   theme_minimal()
```